

Spring Core Master Interview Roadmap

Comprehensive Curriculum & Enterprise Interview Guide

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Module 1: Spring Fundamentals & Architecture

1. What is Spring Framework? (Monolith vs. Modular Architecture)
2. Tight Coupling vs. Loose Coupling Mechanics
3. POJO, POJI, and JavaBean Concepts
4. Spring Framework Core Modules Overview (spring-core, spring-beans, spring-context, spring-aop)

 **Module 1 Mock Interview Included**

Module 2: Inversion of Control (IoC) & Containers

1. What is Inversion of Control (IoC)?
2. Dependency Lookup vs. Dependency Injection
3. IoC Container Architecture: BeanFactory vs. ApplicationContext
4. Lazy Initialization (BeanFactory) vs. Eager Initialization (ApplicationContext)
5. Concrete ApplicationContext Implementations (ClassPathXmlApplicationContext, AnnotationConfigApplicationContext, etc.)

 **Module 2 Mock Interview Included**

Module 3: Dependency Injection (DI) & Configuration Mechanics

1. Setter Injection vs. Constructor Injection vs. Field Injection (Why Field Injection is discouraged)
2. Spring 5 Parameterized Constructor Injection
3. Configuration Styles: XML-Based vs. Annotation-Based vs. Java-Based (@Configuration, @Bean)
4. Externalized Property Management: @PropertySource, @Value("\${...}") vs. SpEL @Value("#{...}")

 **Module 3 Mock Interview Included**

Module 4: Autowiring & Ambiguity Resolution

1. Explicit Wiring vs. Autowiring Modes (byName, byType, constructor)
2. Annotation Autowiring: @Autowired, @Qualifier, @Primary
3. JSR-250 & JSR-330 Standard Annotations: @Resource, @Inject, @Named
4. Handling Multiple Eligible Beans (NoUniqueBeanDefinitionException)
5. Circular Dependency Mechanics & How Spring Solves It (Three-Level Cache Mechanism)

 **Module 4 Mock Interview Included**

Module 5: Bean Scopes & Lifecycle Mechanics

1. Core Bean Scopes: singleton vs. prototype
2. Web-Aware Scopes: request, session, application, websocket
3. Scope Proxy: Injecting Prototype into Singleton (@Scope(proxyMode = ...) / @Lookup)
4. Complete Bean Lifecycle Flow (Instantiation → Aware Callbacks → BeanPostProcessor → Init Callbacks → Destruction)
5. Initialization & Pre-Destroy Annotations (@PostConstruct, @PreDestroy)

 **Module 5 Mock Interview Included**

Module 6: Component Scanning, Profiles & Conditional Configuration

1. Stereotype Annotations: @Component, @Service, @Repository, @Controller
2. @ComponentScan & Custom Include/Exclude Filters
3. Environment & Profiles: @Profile for Multi-Environment Wiring
4. Conditional Bean Registration: @Conditional and Custom Condition Mechanics

 **Module 6 Mock Interview Included**

Module 7: Advanced Spring Core Features

1. Spring Expression Language (SpEL) Essentials
2. Event Handling in Spring: ApplicationEventPublisher and @EventListener
3. Custom BeanPostProcessor & BeanFactoryPostProcessor Use Cases

 **Module 7 Mock Interview Included**

Module 8: Spring Data JDBC & DAO Support

1. JdbcTemplate Core Execution Mechanics
2. Data Operations: update(), queryForObject(), query(), batchUpdate()
3. Mapping Results with RowMapper vs. ResultSetExtractor

 **Module 8 Mock Interview Included**

Module 9: Spring Transaction Management

1. Declarative vs. Programmatic Transactions (TransactionTemplate)
2. PlatformTransactionManager Infrastructure
3. @EnableTransactionManagement & @Transactional Propagation/Isolation Rules

 **Module 9 Mock Interview Included**

Module 10: Spring Aspect-Oriented Programming (AOP)

1. AOP Core Terminology: Aspect, Advice, Join Point, Pointcut, Advisor
2. Proxy Mechanics: JDK Dynamic Proxies vs. CGLIB Proxies
3. Types of Advice: @Before, @After, @AfterReturning, @AfterThrowing, @Around

 **Module 10 Mock Interview Included**

Module 11: Spring Boot Core & Web Fundamentals

1. Spring Boot Auto-Configuration Mechanics (@SpringBootApplication)
2. Web Annotations: @RestController, @Controller, @ResponseBody, @RequestMapping
3. Request Mapping & Parameter Annotations: @PathVariable, @RequestParam, @RequestBody

 **Module 11 Comprehensive Interview Included**



Spring Core vs. Advanced Modules Interview Matrix

Topic Area	Belongs To	Critical Interview Focus Question
Circular Dependency	Spring Core	"How does Spring solve circular dependencies using the 3-level cache?"
Prototype in Singleton	Spring Core	"What happens when you inject a prototype bean into a singleton bean, and how do you fix it?"
Bean Lifecycle	Spring Core	"What is the role of BeanPostProcessor in creating Spring AOP proxies?"
Transactions	Spring Data / TX	"How does @Transactional work internally using Spring AOP proxies?"
Auto-Configuration	Spring Boot	"How does @EnableAutoConfiguration read META-INF / spring.factories?"